Rajarata University of Sri Lanka
Faculty of Applied Sciences
Department of Computing

ICT 1305 – Data Structures

Practical 1: Pointer Basics and Array Manipulation (Duration: ~45 mins)

Objective: Understand pointer declaration, dereferencing, and pointer arithmetic.

Task:

1. Create an integer array of 10 elements.
2. Use a pointer to:
   - Traverse the array.
   - Print all elements.
   - Find and print the sum of elements using pointer arithmetic.
3. Swap the first and last element using pointers.

Learning Outcome:

- Basic pointer operations
- Pointer with arrays
- Swapping via pointers

Practical 2: Pointers with Functions – Swapping and Sorting (Duration: ~1 hour)

Objective: Practice using pointers as function parameters.

Task:

1. Write a function `void swap(int *a, int *b)` to swap two integers.

2. Write a function `void sort(int *arr, int size)` that sorts an array using pointer arithmetic (e.g., Bubble Sort or Selection Sort).

3. In `main`, read `n` integers from the user, sort them using the `sort()` function, and display them.

Bonus: Use the `swap()` function inside the sorting function.

Learning Outcome:

- Pointers as function arguments
- Modular programming with functions
- Practical sorting with pointers

Practical 3: Dynamic Memory Allocation and Pointer-Based String Operations (Duration: ~1 hour 15 mins)

Objective: Practice pointers with dynamic memory and string handling.

Task:

1. Dynamically allocate memory for `n` strings (name list of `n` students).
2. Write a function `void toUpperCase(char *str)` that converts a string to uppercase using pointers.
3. Use the function to convert all names to uppercase.
4. Display all names in uppercase.
5. Free the allocated memory properly.

Learning Outcome:

- Use of `malloc` and `free`
- String manipulation using pointers
- Functions with `char *` parameters

Optional Challenge (If time permits):

- Implement a function `char* concatStrings(char *str1, char *str2)` using dynamic memory allocation and return a new concatenated string.